SIMATS ENGINEERING

CSA17 – ARTIFICIAL INTELLIGENCE LAB

Day 1 – Question 6: 6. Vacuum Cleaner Problem

Aim: To clean two rooms using a simple vacuum-cleaner agent.

Python Program

```
rooms = {'A': 'Dirty', 'B': 'Dirty'}
position = 'A'

while 'Dirty' in rooms.values():
    print("Vacuum at", position)
    if rooms[position] == 'Dirty':
        rooms[position] = 'Clean'
        print(position, "room cleaned")
    else:
        print(position, "room already clean")
    position = 'B' if position == 'A' else 'A'

print("All rooms are clean")
```

Sample Output

```
Vacuum at A
A room cleaned
Vacuum at B
B room cleaned
All rooms are clean
```

Result: The program demonstrates the stated Artificial Intelligence problem.